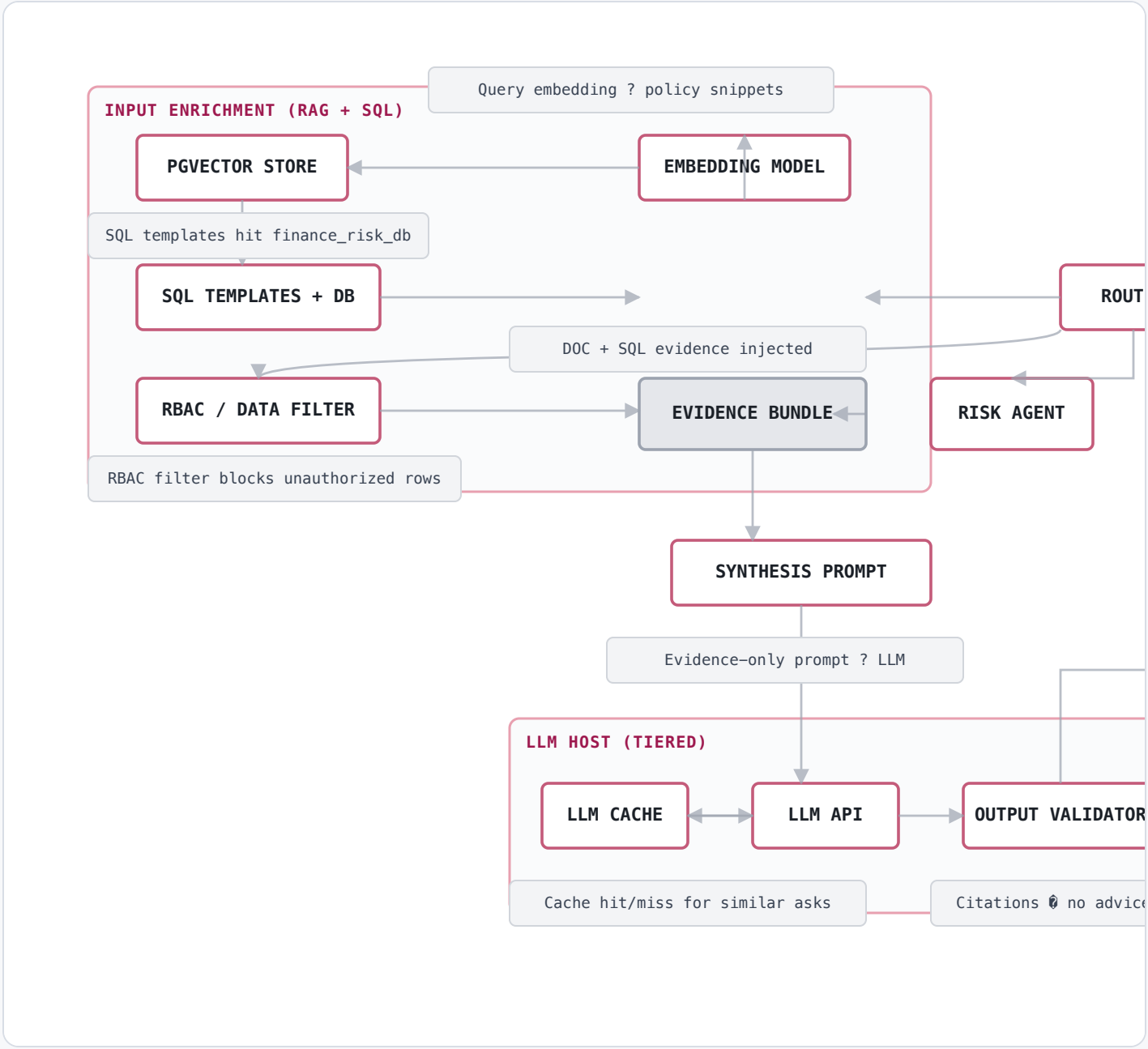
ARCHITECTURE OF THIS LLM APPLICATION FINANCE RISK & INVESTMENT INTELLIGENCE

Light RAG-style request loop adapted from common Architecture of Today s LLM Application diagrams (user ? enrich ? prompt ? model ? safety ? user), customized for our hybrid **RAG + SQL + multi-agent + guardrails** system.

Pause | Prev | Next | Normal | Step 1 of 9

Step 1 of 9: Analyst sends a query
End user submits a natural-language finance/risk question through the UI / FastAPI client.



Components above can be grouped as: **User Input** (END USER, UI/API) **Input Enrichment & Prompt Construction** (embedding, pgvector, SQL, RBAC, evidence bundle, risk, synthesis prompt) **Efficient & Responsible AI Tooling** (LLM host/cache, output validator, escalation, observability).

1 **Analyst sends a query**
End user submits a natural-language finance/risk question through the UI / FastAPI client.

2 **Input guardrail before any model spend**
Override / bypass / injection intent is blocked. Clean requests continue to the Router.

3 **RAG enrichment embed & retrieve policy**
Copy of the query is embedded and mapped to pgvector policy_chunks (7 PDFs, page/section metadata).

4 **SQL enrichment templated DB lookup**
For C2/C3/C4, named parameterized SQL templates read finance_risk_db (portfolios, risk logs,).

5 **RBAC / data filter**
Every result set is scoped to allowed_portfolio_ids. Unauthorized data never reaches the prompt.

6 **Evidence bundle + risk reconcile**
DOC and SQL snippets converge into the Evidence Bundle. Risk Agent scores / detects breaches.

7 **Synthesis prompt ? LLM host**
Tool-less synthesis prompt is built from evidence only, then sent to the tiered LLM API (with optional cache).

8 **Output validator & escalation gates**
Citations, non-advice, disclaimer checked. Any E1E8 trigger becomes a human-handoff envelope instead of an answer.

9 **Answer returns to the analyst**
Grounded response (or escalation refusal) flows back through UI/API to the end user, with full telemetry.